



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2

Division: A

Subject: Programming with Java

ASSIGNMENT No : 19

TITLE : Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory

Java Database Connectivity (**JDBC**) is a standard Java API that enables communication between Java applications and relational databases. It provides a platform-independent mechanism to connect to databases such as MySQL, Oracle, PostgreSQL, and SQL Server. Before executing SQL statements, a JDBC driver is loaded and a connection is established using the `DriverManager` class. The `Connection` object represents an active connection to the database, while the `Statement` object is used to execute SQL queries. The `SELECT` statement retrieves records from database tables based on specified conditions. The retrieved records are stored in a `ResultSet` object, which allows sequential access to the data. JDBC simplifies database programming and is widely used in desktop, web, and enterprise applications.

Output Screenshots

```
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp19/Main.java
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp19.Main
1 Devansh BCA
2 Rahul BCA
3 Aman BCA
❖ dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE 1

```
1 package Exp19;
2
3 import java.sql.*;
4
5 public class Main {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college",
12                "root",
13                "password"
14            );
15
16            Statement stmt = con.createStatement();
17
18            ResultSet rs = stmt.executeQuery("SELECT * FROM
student");
19
20            while (rs.next()) {
21                System.out.println(
22                    rs.getInt("id") + " " +
23                    rs.getString("name") + " " +
24                    rs.getString("course")
25                );
26            }
27
28            con.close();
29        }
30        catch (Exception e) {
31            System.out.println(e);
32        }
33    }
34 }
```

```
32     }
33 }
34 }
```

EXERCISE 1

```
1 package Exp19.Exercise;
2
3 import java.sql.*;
4
5 public class Exp1 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college",
12                "root",
13                "password"
14            );
15
16            Statement stmt = con.createStatement();
17
18            ResultSet rs = stmt.executeQuery(
19                "SELECT * FROM student"
20            );
21
22            System.out.println("Student Records:");
23
24            while (rs.next()) {
25                System.out.println(
26                    "ID: " + rs.getInt("id") +
27                    ", Name: " + rs.getString("name") +
28                    ", Course: " + rs.getString("course")
29                );
30            }
31
32            con.close();
33        }
34        catch (Exception e) {
35            System.out.println("Error: " + e.getMessage());
36        }
37    }
38 }
```

```
36     }
37 }
38 }
```

EXERCISE 2

```
1 package Exp19.Exercise;
2
3 import java.sql.*;
4
5 public class Exp2 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/shop",
12                "root",
13                "password"
14            );
15
16            Statement stmt = con.createStatement();
17
18            ResultSet rs = stmt.executeQuery(
19                "SELECT * FROM product"
20            );
21
22            System.out.println("Product Details:");
23
24            while (rs.next()) {
25                System.out.println(
26                    "Product ID: " + rs.getInt("product_id") +
27                    ", Name: " + rs.getString("product_name") +
28                    ", Quantity: " + rs.getInt("quantity") +
29                    ", Price: " + rs.getDouble("price")
30                );
31            }
32
33            con.close();
34        }
```

```
35         catch (Exception e) {
36             System.out.println("Error: " + e.getMessage());
37         }
38     }
39 }
```

SQL CODE:

```
1 CREATE DATABASE college;
2 CREATE DATABASE shop;
3
4 USE college;
5
6 CREATE TABLE student (
7     id INT PRIMARY KEY,
8     name VARCHAR(50),
9     course VARCHAR(50)
10 );
11
12 INSERT INTO student VALUES
13 (1, 'Devansh', 'BCA'),
14 (2, 'Rahul', 'BCA'),
15 (3, 'Aman', 'BCA');
16
17 USE shop;
18
19 CREATE TABLE product (
20     product_id INT PRIMARY KEY,
21     product_name VARCHAR(50),
22     quantity INT,
23     price DOUBLE
24 );
25
26 INSERT INTO product VALUES
27 (101, 'Laptop', 5, 55000),
28 (102, 'Mouse', 20, 700),
29 (103, 'Keyboard', 10, 1200);
```

Exercise

Q1. Create a Java application to display student records from database using SELECT query.

```
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp19/Exercise/Exp1.java
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp19.Exercise.Exp1
Student Records:
ID: 1, Name: Devansh, Course: BCA
ID: 2, Name: Rahul, Course: BCA
ID: 3, Name: Aman, Course: BCA
❖ dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

```
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp19/Exercise/Exp2.java
● dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp19.Exercise.Exp2
Product Details:
Product ID: 101, Name: Laptop, Quantity: 5, Price: 55000.0
Product ID: 102, Name: Mouse, Quantity: 20, Price: 700.0
Product ID: 103, Name: Keyboard, Quantity: 10, Price: 1200.0
❖ dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

Thus, the experiment successfully demonstrated the use of JDBC for establishing connectivity between a Java application and a relational database. The program used JDBC components such as DriverManager, Connection, Statement, and ResultSet to establish a connection, execute a SELECT query, retrieve records, and display the required data. This experiment provided an understanding of how Java applications can interact with databases and perform basic data retrieval operations efficiently.